

Volume II · Chapter 10 — Benchmarking & Hallucination · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-10-context.md for the AI interaction log.

Prerequisites: Ch 8, Ch 9.

Learning outcomes — the student can: evaluate LLMs on medical benchmarks (MedQA/PubMedQA/MedMCQA); interpret results; measure hallucination; build an eval harness; explain why benchmark $\neq$ clinical safety.

Topics (theory) — 8 h:

#	Topic	h
10.1	Benchmark landscape: MedQA (USMLE), PubMedQA, MedMCQA	2
10.2	Metrics: accuracy, exact-match & their limits	1
10.3	Hallucination: definitions, causes, measurement	2
10.4	Faithfulness & grounding metrics	1
10.5	Benchmark $\neq$ clinical safety; distribution shift	2

Datasets/tools: HF datasets/evaluate; local LLM. **Assessment:** eval-harness deliverable (50%); hallucination report (30%); quiz (20%). **Key decisions:** which benchmark supports which claim; honest reporting of limitations. **References:** plan §13 → *Fine-tuning, alignment & benchmarks; Medical LLMs*. **Hours:** Theory 8 + Lab 8 = 16.